

➤ **Topic: Basic Linux**

➤ **Objective of Linux:** The goal of a Linux Command is to perform a specific task such as managing files, users, processes or the system.

➤ **Purpose:** It helps users interact with the operating system efficiently through the command line.

➤ **Usage of Commands**

➤ **echo \$0:** It Displays the name of the current terminal.

➤ **pwd:** It shows the current working directory and helps the location where the Linux file exist.

➤ **/(Forward slash):** It always represents the root directory.

➤ **~(Tild):** It always represents the home directory.

➤ **cd:** It is used to change the directory.

➤ **cd . :** It is used to move from current directory to its previous directory.

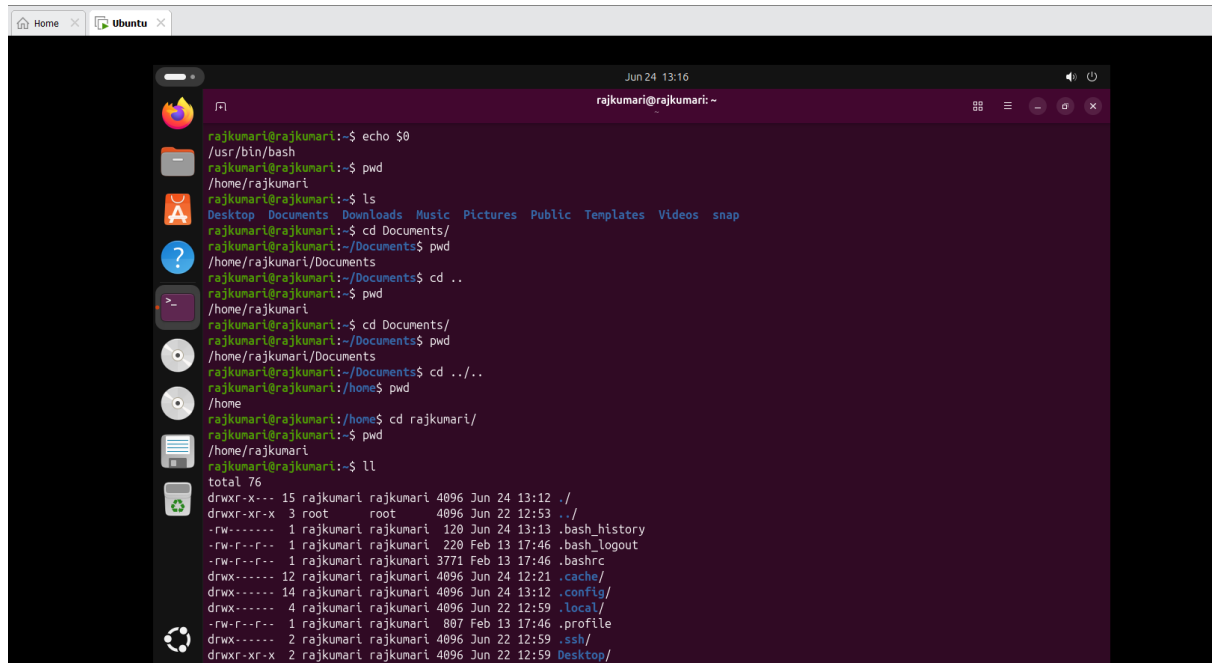
➤ **cd ../.. :** It is used to move from current directory to two previous directories.

➤ **cd~ :** Directly navigate to the current user's home directly.

➤ **cd/ :** Directly navigate to the root directory of the Linux file system.

➤ **ls :** It is list out the all the files and directories in the current directory.

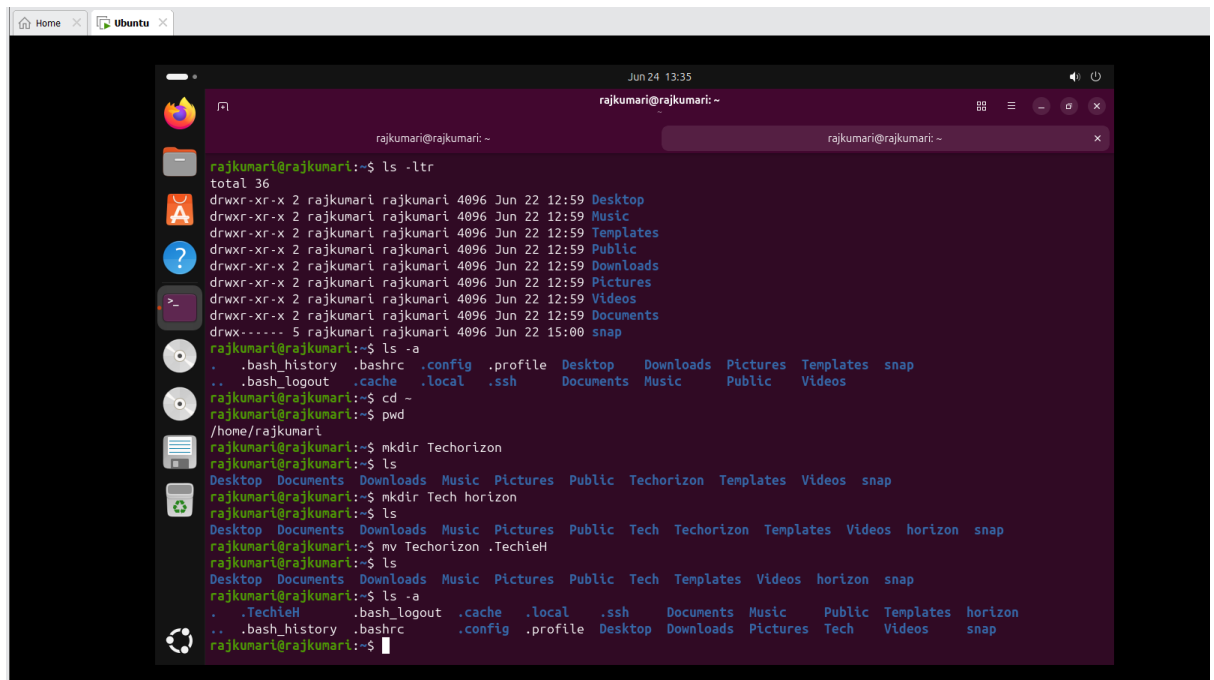
- **ll** : It is list out the all the files and directories in a detailed format like owner, size, date and file name.



The screenshot shows a terminal window with a dark purple background. The user is logged in as 'rajkumari' on a machine named 'rajkumari'. The terminal shows the following commands and their outputs:

```
rajkumari@rajkumari:~$ echo $0
/usr/bin/bash
rajkumari@rajkumari:~$ pwd
/home/rajkumari
rajkumari@rajkumari:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snap
rajkumari@rajkumari:~$ cd Documents/
rajkumari@rajkumari:~/Documents$ pwd
/home/rajkumari/Documents
rajkumari@rajkumari:~/Documents$ cd ..
rajkumari@rajkumari:~$ pwd
/home/rajkumari
rajkumari@rajkumari:~$ cd Documents/
rajkumari@rajkumari:~/Documents$ pwd
/home/rajkumari/Documents
rajkumari@rajkumari:~/Documents$ cd ../../
rajkumari@rajkumari:~$ pwd
/home
rajkumari@rajkumari:~$ cd rajkumari/
rajkumari@rajkumari:~/rajkumari$ pwd
/home/rajkumari
rajkumari@rajkumari:~/rajkumari$ ll
total 76
drwxr-x--- 15 rajkumari rajkumari 4096 Jun 24 13:12 ./
drwxr-xr-x  3 root      root      4096 Jun 22 12:53 ../
-rw-r----- 1 rajkumari rajkumari 120 Jun 24 13:13 .bash_history
-rw-r----- 1 rajkumari rajkumari 220 Feb 13 17:46 .bash_logout
-rw-r----- 1 rajkumari rajkumari 3771 Feb 13 17:46 .bashrc
drwx----- 12 rajkumari rajkumari 4096 Jun 24 12:21 .cache/
drwx----- 14 rajkumari rajkumari 4096 Jun 24 13:12 .config/
drwx----- 4 rajkumari rajkumari 4096 Jun 22 12:59 .local/
-rw-r----- 1 rajkumari rajkumari 807 Feb 13 17:46 .profile
drwx----- 2 rajkumari rajkumari 4096 Jun 22 12:59 .ssh/
drwxr-xr-x  2 rajkumari rajkumari 4096 Jun 22 12:59 Desktop/
```

- **ls -ltr**: It is used to list out the files and directories in detailed format, sorted by modification time.
- **-l** → long listing format (detailed information)
- **-t** → sort by modification time
- **-r** → Reverse the order (oldest files first)
- **ls -a** : list out all the files and directories, including hidden files.
- **ls -la** : list out all the files , directories and hidden files in detailed format.
- **ls -ls** : list the files in sorting format.
- **ls -d**: list all the directories only instead of display their content.
- **mkdir**: It helps to create a new directory(folder).
- **mv**: It helps to move or rename the files and directories.



```
rajkumari@rajkumari:~$ ls -ltr
total 36
drwxr-xr-x 2 rajkumari rajkumari 4096 Jun 22 12:59 Desktop
drwxr-xr-x 2 rajkumari rajkumari 4096 Jun 22 12:59 Music
drwxr-xr-x 2 rajkumari rajkumari 4096 Jun 22 12:59 Templates
drwxr-xr-x 2 rajkumari rajkumari 4096 Jun 22 12:59 Public
drwxr-xr-x 2 rajkumari rajkumari 4096 Jun 22 12:59 Downloads
drwxr-xr-x 2 rajkumari rajkumari 4096 Jun 22 12:59 Pictures
drwxr-xr-x 2 rajkumari rajkumari 4096 Jun 22 12:59 Videos
drwxr-xr-x 2 rajkumari rajkumari 4096 Jun 22 12:59 Documents
drwx----- 5 rajkumari rajkumari 4096 Jun 22 15:00 snap

rajkumari@rajkumari:~$ ls -la
. .bash_history .bashrc .config .profile Desktop Downloads Pictures Templates snap
.. .bash_logout .cache .local .ssh Documents Music Public Videos

rajkumari@rajkumari:~$ cd ~
rajkumari@rajkumari:~$ pwd
/home/rajkumari

rajkumari@rajkumari:~$ mkdir Techorizon
rajkumari@rajkumari:~$ ls
Desktop Documents Downloads Music Pictures Public Techorizon Templates Videos snap

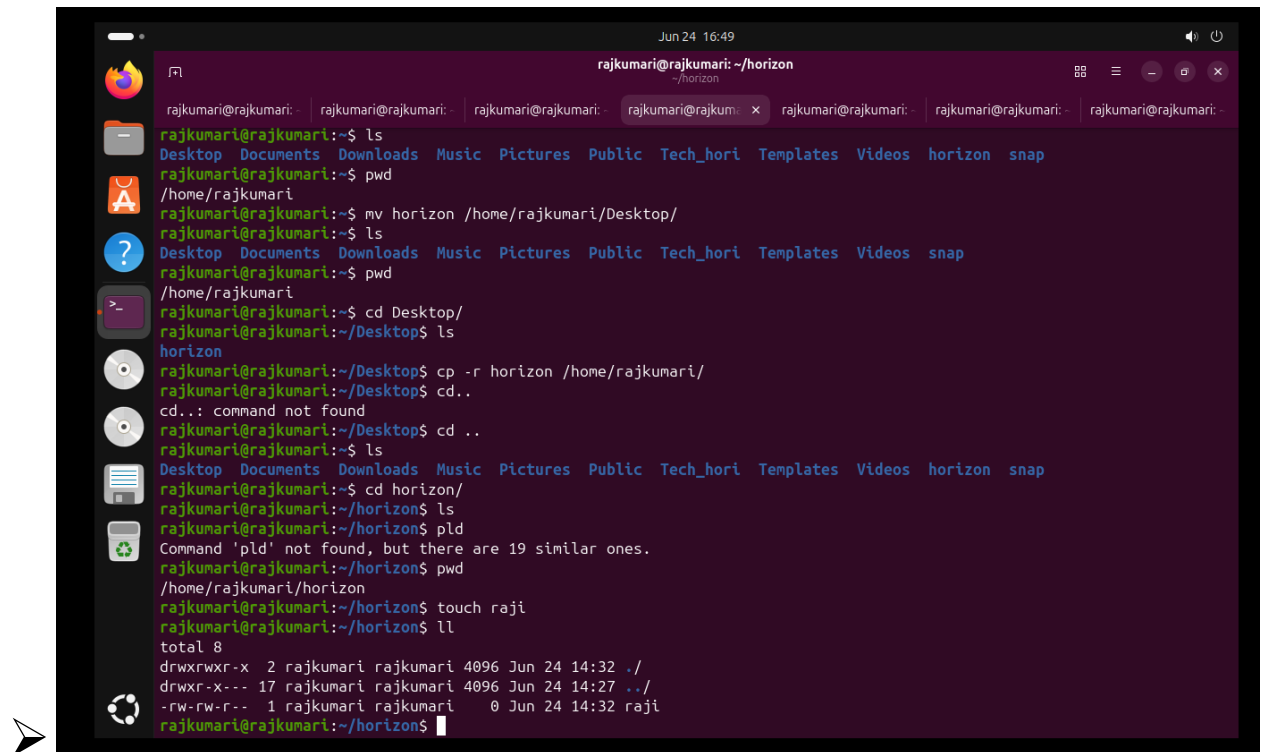
rajkumari@rajkumari:~$ mkdir Tech horizon
rajkumari@rajkumari:~$ ls
Desktop Documents Downloads Music Pictures Public Tech Techorizon Templates Videos horizon snap

rajkumari@rajkumari:~$ mv Techorizon .TechieH
rajkumari@rajkumari:~$ ls
Desktop Documents Downloads Music Pictures Public Tech Templates Videos horizon snap

rajkumari@rajkumari:~$ ls -la
. .TechieH .bash_logout .cache .local .ssh Documents Music Public Templates horizon
.. .bash_history .bashrc .config .profile Desktop Downloads Pictures Tech Videos snap

rajkumari@rajkumari:~$
```

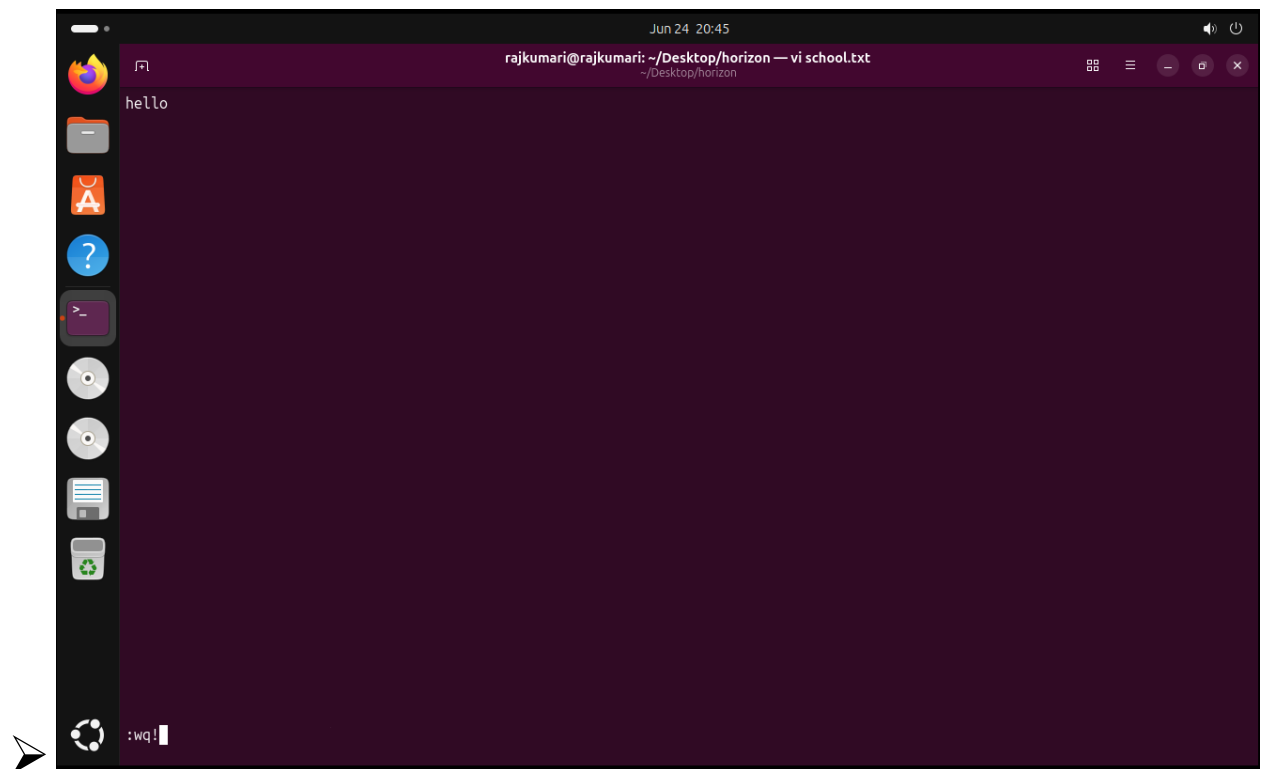
- **cp**: Copy files or directories from one location to another.
- **cp -r**: copy a directory and all its contents, including subdirectories and files.
- **touch**: Create a new empty file or update the timestamp of an existing file.



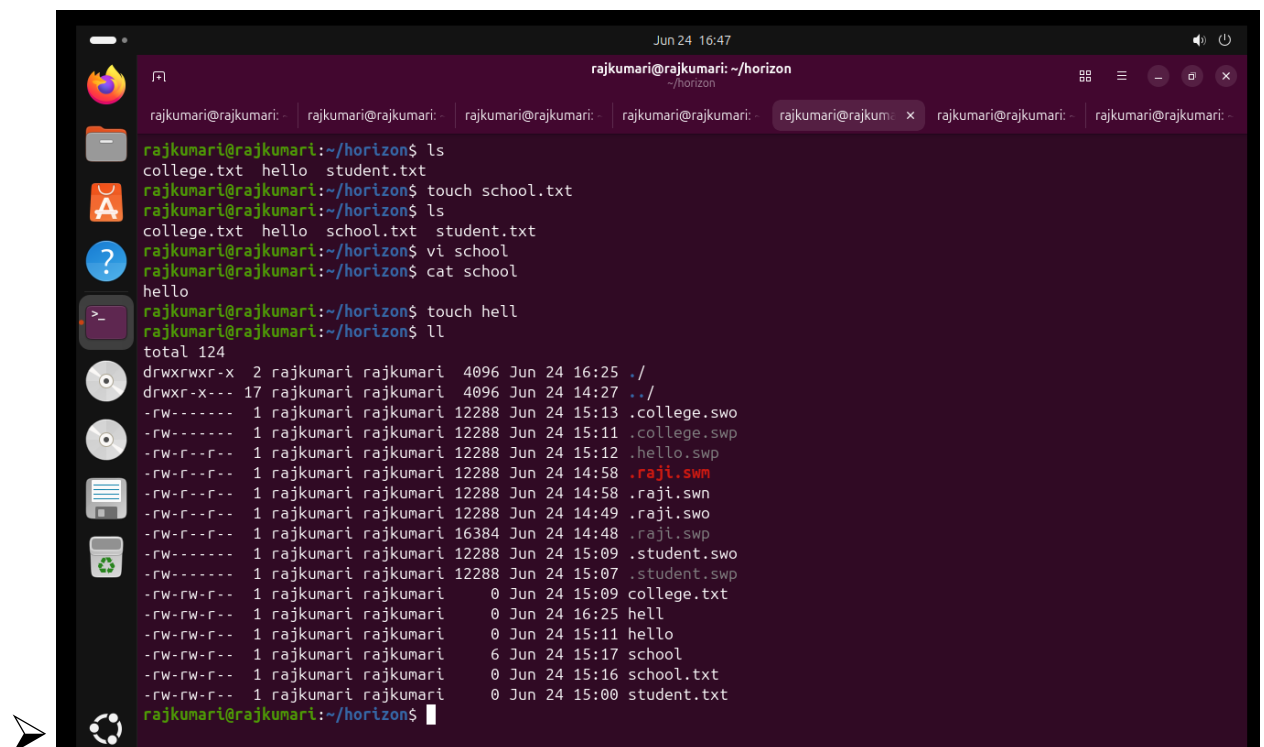
The screenshot shows a terminal window with a dark purple background. The user is performing several file operations: listing files, moving a directory, copying a directory, and creating a new file. The terminal output is as follows:

```
rajkumari@rajkumari: ~$ ls
Desktop Documents Downloads Music Pictures Public Tech_hori Templates Videos horizon snap
rajkumari@rajkumari: ~$ pwd
/home/rajkumari
rajkumari@rajkumari: ~$ mv horizon /home/rajkumari/Desktop/
rajkumari@rajkumari: ~$ ls
Desktop Documents Downloads Music Pictures Public Tech_hori Templates Videos snap
rajkumari@rajkumari: ~$ pwd
/home/rajkumari
rajkumari@rajkumari: ~$ cd Desktop/
rajkumari@rajkumari: ~/Desktop$ ls
horizon
rajkumari@rajkumari: ~/Desktop$ cp -r horizon /home/rajkumari/
rajkumari@rajkumari: ~/Desktop$ cd..
cd..: command not found
rajkumari@rajkumari: ~/Desktop$ cd ..
rajkumari@rajkumari: ~$ ls
Desktop Documents Downloads Music Pictures Public Tech_hori Templates Videos horizon snap
rajkumari@rajkumari: ~$ cd horizon/
rajkumari@rajkumari: ~/horizon$ ls
rajkumari@rajkumari: ~/horizon$ pld
Command 'pld' not found, but there are 19 similar ones.
rajkumari@rajkumari: ~/horizon$ pwd
/home/rajkumari/horizon
rajkumari@rajkumari: ~/horizon$ touch raji
rajkumari@rajkumari: ~/horizon$ ll
total 8
drwxrwxr-x  2 rajkumari rajkumari 4096 Jun 24 14:32 ./
drwxr-x--- 17 rajkumari rajkumari 4096 Jun 24 14:27 ../
-rw-rw-r--  1 rajkumari rajkumari   0 Jun 24 14:32 raji
rajkumari@rajkumari: ~/horizon$
```

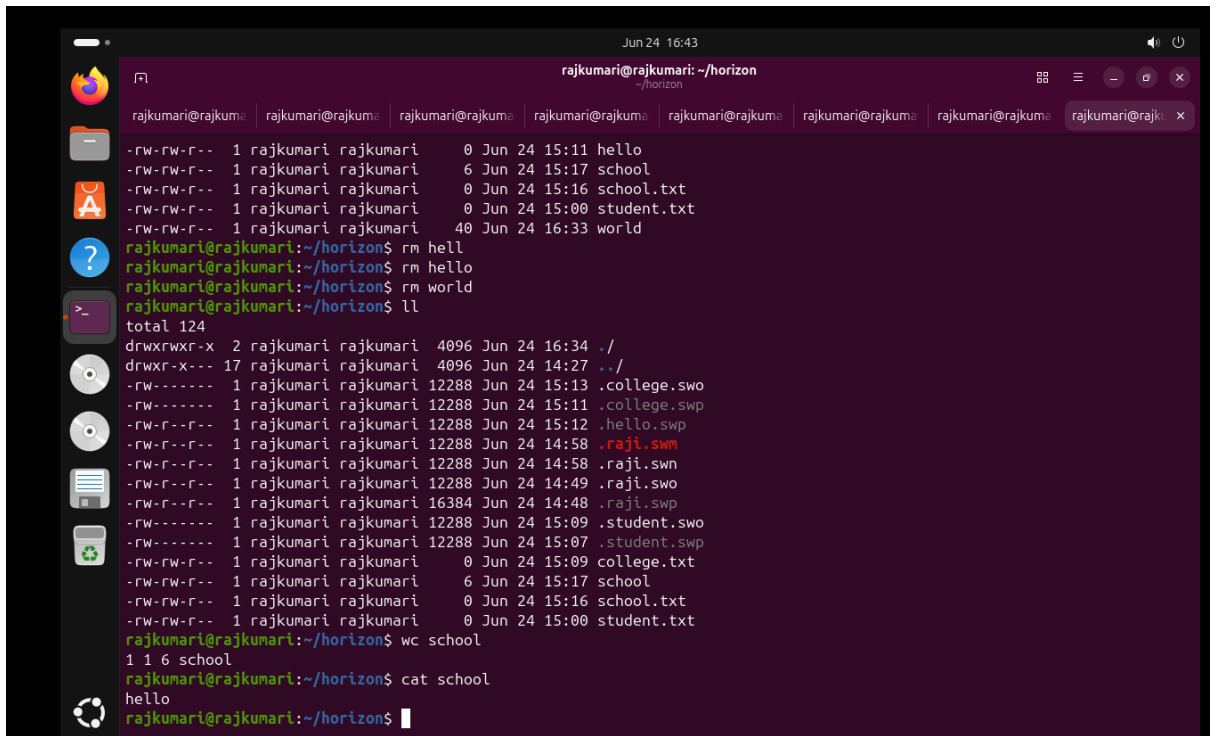
- **vi**: vi is the text editor. It is used to edit the text file.
- Press ‘**i**’ : To enter Insert mode and type text.
- Press ‘**Esc**’ : To exit the insert mode.
- Type “**:wq!**” : To save and quit from file.



➤ **cat:** It is used to display the content of a file.



- **rm**: Remove(delete) files or directories.
- **rm -rf**: Delete a directory and all its contents recursively without asking for confirmation.
- **rmdir**: Remove an empty directory.
- **wc**: It is used to count the lines, words, and characters in a file.



```

Jun 24 16:43
rajkumari@rajkumari: ~/horizon
rajkumari@rajkumari: ~/horizon$ ll
total 124
drwxrwxr-x 2 rajkumari rajkumari 4096 Jun 24 16:34 ./
drwxr-x--- 17 rajkumari rajkumari 4096 Jun 24 14:27 ../
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:13 .college.swo
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:11 .college.swp
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:12 .hello.swp
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 14:58 .raji.swm
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 14:58 .raji.swn
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 14:49 .raji.swo
-rw-r--r-- 1 rajkumari rajkumari 16384 Jun 24 14:48 .raji.swp
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:09 .student.swo
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:07 .student.swp
-rw-r--r-- 1 rajkumari rajkumari 0 Jun 24 15:09 college.txt
-rw-r--r-- 1 rajkumari rajkumari 6 Jun 24 15:17 school
-rw-r--r-- 1 rajkumari rajkumari 0 Jun 24 15:16 school.txt
-rw-r--r-- 1 rajkumari rajkumari 0 Jun 24 15:00 student.txt
rajkumari@rajkumari:~/horizon$ rm hell
rajkumari@rajkumari:~/horizon$ rm hello
rajkumari@rajkumari:~/horizon$ rm world
rajkumari@rajkumari:~/horizon$ ll
total 124
drwxrwxr-x 2 rajkumari rajkumari 4096 Jun 24 16:34 ./
drwxr-x--- 17 rajkumari rajkumari 4096 Jun 24 14:27 ../
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:13 .college.swo
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:11 .college.swp
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:12 .hello.swp
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 14:58 .raji.swm
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 14:58 .raji.swn
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 14:49 .raji.swo
-rw-r--r-- 1 rajkumari rajkumari 16384 Jun 24 14:48 .raji.swp
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:09 .student.swo
-rw-r--r-- 1 rajkumari rajkumari 12288 Jun 24 15:07 .student.swp
-rw-r--r-- 1 rajkumari rajkumari 0 Jun 24 15:09 college.txt
-rw-r--r-- 1 rajkumari rajkumari 6 Jun 24 15:17 school
-rw-r--r-- 1 rajkumari rajkumari 0 Jun 24 15:16 school.txt
-rw-r--r-- 1 rajkumari rajkumari 0 Jun 24 15:00 student.txt
rajkumari@rajkumari:~/horizon$ wc school
1 1 6 school
rajkumari@rajkumari:~/horizon$ cat school
hello
rajkumari@rajkumari:~/horizon$

```

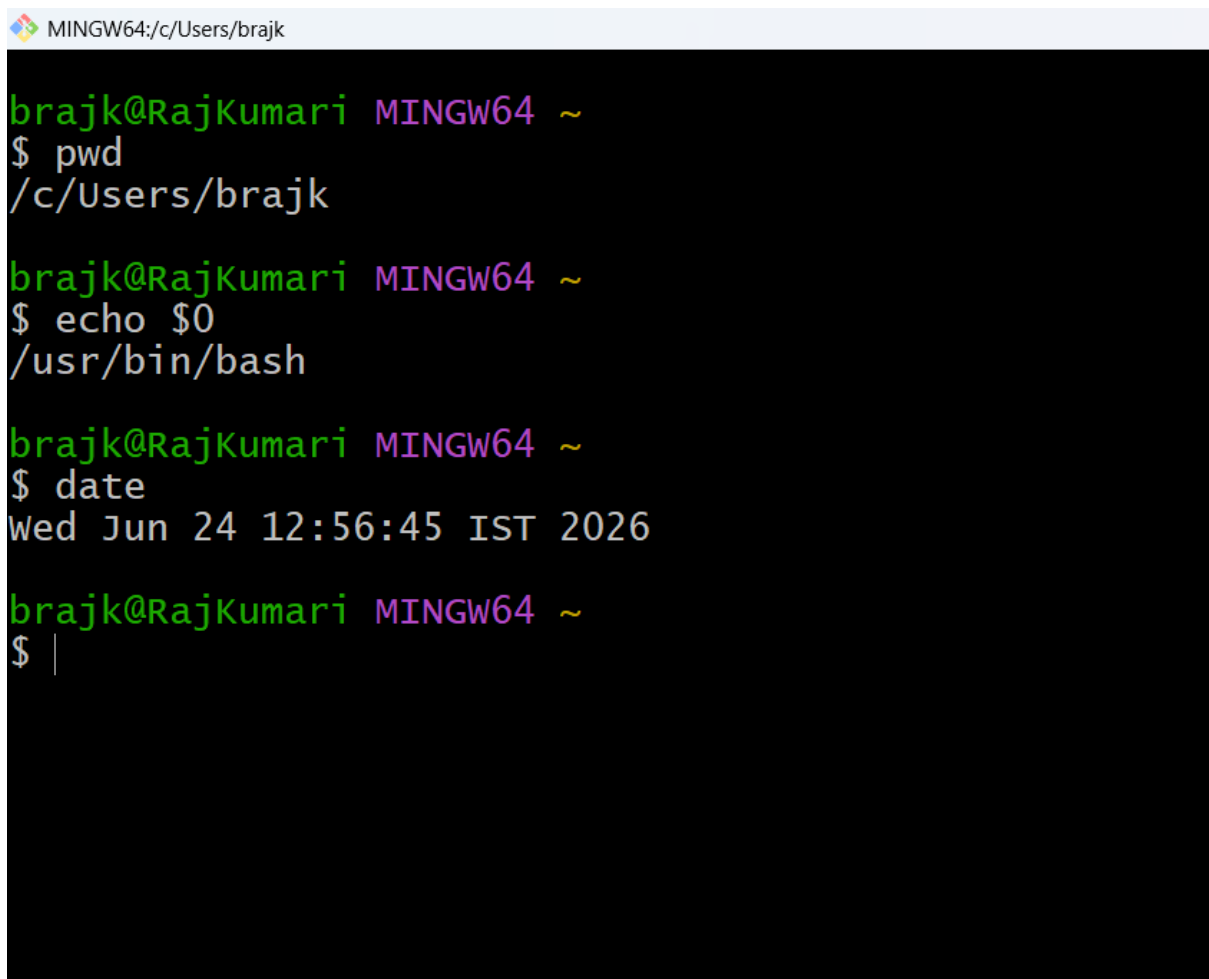
- **date**: It Display the current system date and time.
- **history**: it display the list of commands executed previously.

```
Jun 24 16:50
rajkumari@rajkumari: ~
rajkumari@rajkumari:~$ ls
Desktop Documents Downloads Music Pictures Public Tech Tech_hori Templates Videos horizon snap
rajkumari@rajkumari:~$ history
1 echo $0
2 ls
3 cd /
4 pwd
5 cd -
6 pwd
7 cd Desktop/
8 pwd
9 cd ..
10 pwd
11 cd Desktop/
12 cd Documents/
13 pwd
14 cd..
15 cd ..
16 pwd
17 ll
18 echo $0
19 ll
20 pwd
21 ls
22 history
rajkumari@rajkumari:~$ date
Wed Jun 24 01:42:41 PM IST 2026
rajkumari@rajkumari:~$ ls
Desktop Documents Downloads Music Pictures Public Tech Tech_hori Templates Videos horizon snap
rajkumari@rajkumari:~$
```

```
Jun 24 16:54
rajkumari@rajku: ~
rajkumari@rajku:~/horizon$ cd ..
rajkumari@rajku:~$ pwd
/home/rajkumari
rajkumari@rajku:~$ history
1 echo $0
2 ls
3 cd /
4 pwd
5 cd -
6 pwd
7 cd Desktop/
8 pwd
9 cd ..
10 pwd
11 cd Desktop/
12 cd Documents/
13 pwd
14 cd..
15 cd ..
16 pwd
17 ll
18 echo $0
19 ll
20 pwd
21 vi raji
22 vi raji
23 VI raji
24 vi raji
25 ls
26 cd horizon/
```

```
Jun 24 17:12
rajkumari@rajkumari: ~
rajkumari@rajku: ~
32 vi student
33 cat student
34 ls
35 rm -rf raji
36 ls
37 rm -rf student
38 ls
39 vi student
40 ls
41 vi student
42 ls
43 vi student
44 ls
45 touch college.txt
46 ld
47 ls
48 vi college
49 touch hello
50 ls
51 vi hello
52 ls
53 vi college
54 ls
55 cat school
56 touch world
57 ll
58 cd ..
59 pwd
60 history
rajkumari@rajkumari:~$
```

- **gitbash:** It is a terminal application for Windows that provides a Linux-like command line environment and Git commands.
- **Objective:** Provide a Linux-style terminal on Windows.
- **Purpose:** Run Git and Linux commands easily.
- **Command Tool:** Git Bash combines Git functionality with a Bash shell environment.



The screenshot shows a Windows terminal window titled "MINGW64:/c/Users/brajk". The terminal has a black background with green and purple text. It displays four command-line interactions:

```
brajk@RajKumari MINGW64 ~  
$ pwd  
/c/Users/brajk  
  
brajk@RajKumari MINGW64 ~  
$ echo $0  
/usr/bin/bash  
  
brajk@RajKumari MINGW64 ~  
$ date  
Wed Jun 24 12:56:45 IST 2026  
  
brajk@RajKumari MINGW64 ~  
$ |
```